

OPM-5020 Project

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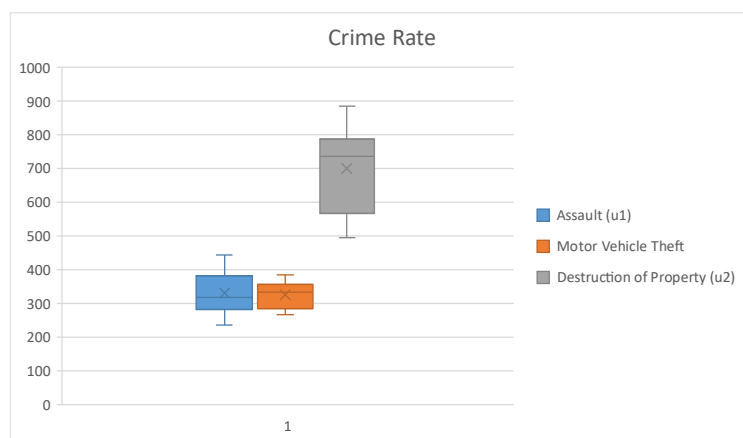
For our group project our team started out by formulating a hypothesis, wrapped around a couple of assumptions and began to search for supporting data to help us derive a conclusion. What our group has decided to explore was if we could determine which individual crime in Saint Louis City had the greatest impact on the overall crime rating for the city.

To help run this test we pulled data from the Saint Louis Metropolitan Police Department National Incident-Based Reporting System (NIBRS) for 2021. For the NIBRS we could determine and pull the total crimes reported in Saint Louis. Additionally, we also pulled 2021 data from the Census Bureau for the population of the city to help us determine the monthly crime rate.

Suppose you wanted to understand what all went into formulating the total crime rate of an area. One further you wanted to know which of the individual crimes had a greater influence on that crime rating. In the analysis we performed we did just that. We first investigated what was used to establish the crime rating for Saint Louis City. Next, we calculated the total number of crimes in 2021 by pulling the data from the

NIBRS and dividing that number by the city's population in 2021 and multiplied that by 100 to get our (y) total crime rate.

So, from the following box and whisker graph we can determine that the median values for assault, motor vehicle theft and destruction of property are 316.5, 332.5, 736 respectively.



We also selected multiple other variables, or individual crimes that were used to see how they individually influenced the crime rating. We began by selecting 3 independent variables Assault, Motor Vehicle Theft, and Destruction of Property (DoP). Ultimately, what we focused our analysis on was the relationship of Assault on the crime rate and Destruction of Property on the crime rate.

For our hypothesis we used mean of assault as μ_1 , motor Vehicle Theft is μ_2 , and the mean of Destruction if Property is μ_3 . As we can see that μ_3 is greater than μ_2 and μ_1 and it

affects the crime rate the most among these variables which we can see in the R console regression. Thus, we are checking whether μ_1 is greater than μ_2 i.e., which contributes more to the crime rate.

From the regression analysis, by looking at the R square we can say that approximately 94% of the values can be explained in this model, also that the Ttest falls outside the rejection region additionally, the p-value is greater than the alpha so same conclusion. Hence there is not sufficient evidence to reject the null hypothesis which means that $\mu_1 = \mu_2$ which is both the variables (Assault and Motor Vehicle Theft) are contributing almost equally to the crime rate and the model is also significant since the significance F is too low.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.97187537							
R Square	0.94454173							
Adjusted R Squ	0.92374488							
Standard Error	0.04961037							
Observations	12							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	3	0.33534332	0.11178111	45.4175334	2.27566E-05			
Residual	8	0.01968951	0.00246119					
Total	11	0.35503283						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.31750322	0.13401877	2.36909517	0.04531037	0.008455383	0.62655106	0.00845538	0.62655106
Assault	0.00073602	0.00037082	1.98481964	0.08243148	-0.000119102	0.00159113	-0.0001191	0.00159113
Motor Vehicle	0.00134553	0.00050236	2.67841856	0.02799453	0.000187085	0.00250396	0.00018709	0.00250396
Destruction of	0.0008537	0.00013844	6.16659384	0.00026906	0.000534459	0.00117294	0.00053446	0.00117294

In addition, we performed a regression analysis in R to determine the relationship between destruction of property and the crime rate and what we observed here from the R square that 77% of the data can be explained from this model and the p-value is <0.05 therefore we can confirm there is a linear relationship.

```
> lmrate <- lm(DOP ~ CRIMERATE, data=dfrate)
> summary(lmrate)
```

Call:
lm(formula = DOP ~ CRIMERATE, data = dfrate)

Residuals:

Min	1Q	Median	3Q	Max
-86.436	-28.722	-6.658	3.639	153.639

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-339.6	180.0	-1.886	0.088624 .
CRIMERATE	651.6	112.2	5.808	0.000171 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 66.85 on 10 degrees of freedom
Multiple R-squared: 0.7713, Adjusted R-squared: 0.7485
F-statistic: 33.73 on 1 and 10 DF, p-value: 0.0001711

```
> cor(dfrate)
```

	CRIMERATE	Assault	DOP
CRIMERATE	1.0000000	0.7988106	0.8782574
Assault	0.7988106	1.0000000	0.5850676
DOP	0.8782574	0.5850676	1.0000000

So, as we can see above we can determine that with a .8782574 correlation destruction of property is closer to 1. What we can conclusively determine from our regression model ran in R is when compared to destruction of property the other tested independent variables tested destruction of property by far has a greater impact on the total crime rate. Again, this applies the previously stated assumption that assault and motor

vehicle theft are equal and therefore will provide a similar Influence. In conclusion we agree that assault and Motor vehicle theft relatively have an equal influence on the total crime rate.

References

SLMPD NIBRS crime statistics. (n.d.). Retrieved April 8, 2023, from https://www.slmpd.org/crime_stats.shtml

Crime insight. (n.d.). Retrieved April 11, 2023, from <https://showmecrime.mo.gov/CrimeReporting/CrimeReportingTOPS.html>

U.S. Census Bureau Quickfacts: St. Louis City, Missouri.
<https://www.census.gov/quickfacts/stlouiscitymissouri>.

References below are for rejected data.

US Department of Commerce, N. O. A. A. (2015, July 10). *Monthly climate data*. National Weather Service. Retrieved April 8, 2023, from <https://www.weather.gov/lx/monthTab>

Current US inflation rates: 2000-2023. US Inflation Calculator | Easily calculate how the buying power of the U.S. dollar has changed from 1913 to 2023. Get inflation rates and U.S. inflation news. (2023, April 4). Retrieved April 8, 2023, from <https://www.usinflationcalculator.com/inflation/current-inflation-rates/>

Published by Statista Research Department, & 16, M. (2023, March 16). *Monthly inflation rate U.S. 2023*. Statista. Retrieved April 8, 2023, from <https://www.statista.com/statistics/273418/unadjusted-monthly-inflation-rate-in-the-us/>

R-Studio

```
> boxplot(dfrate$DOP)
> qqnorm(dfrate$DOP)
> shapiro.test(dfrate$DOP)
```

Shapiro-Wilk normality test

```
data: dfrate$DOP
W = 0.90723, p-value = 0.1965
```

```
> lmrate <- lm(DOP ~ CRIMERATE, data=dfrate)
> summary(lmrate)
```

Call:

```
lm(formula = DOP ~ CRIMERATE, data = dfrate)
```

Residuals:

Min	1Q	Median	3Q	Max
-86.436	-28.722	-6.658	3.639	153.639

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-339.6	180.0	-1.886	0.088624 .
CRIMERATE	651.6	112.2	5.808	0.000171 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 66.85 on 10 degrees of freedom

Multiple R-squared: 0.7713, Adjusted R-squared: 0.7485

F-statistic: 33.73 on 1 and 10 DF, p-value: 0.0001711

```
> dfTestCRIMERATE = data.frame(CRIMERATE=1.55)
> predict(lmrate, dfTestCRIMERATE)
1
670.4366
> predict(lmrate, dfTestCRIMERATE, interval="confidence", level=.90)
      fit      lwr      upr
1 670.4366 634.256 706.6171
> predict(lmrate, dfTestCRIMERATE, interval="prediction", level=.90)
```

```

      fit      lwr      upr
1 670.4366 543.9838 796.8894
> plot (dfrate$CRIMRATE, dfrate$DOP, main="Regress DOP on CRIMRATE",
+       xlab="DOP", ylab="CRIMRATE 000s", frame=FALSE)
> abline(lmrate, col="blue")

> cor(dfrate)
      CRIMRATE  Assault      DOP
CRIMRATE 1.0000000 0.7988106 0.8782574
Assault   0.7988106 1.0000000 0.5850676
DOP       0.8782574 0.5850676 1.0000000

```